9	A stone is attached to a string. The stone is then caused to swing in a vertical circular motion at a constant speed. Which of the following statements is ***incorrect***?	
	A	The magnitude of resultant force acting on the stone is constant throughout the circular motion.
	B	The acceleration is always directed towards the centre of the circle throughout the circular motion.
	C	The kinetic energy of the stone is constant throughout the circular motion.
	D	The tension in the string when the stone is at the highest point of the circular motion is higher than that when the stone is at the lowest point.
	Answer: D Centripetal force is constant in magnitude The centripetal acceleration is always acting towards the centre of circle K.E. is constant because speed is constant since KE = ½ m v^2 At lowest point, tension – weight = centripetal force Tension = centripetal force + weight At highest point, tension + weight = centripetal force Tension = centripetal force - weight Tension at the highest point is lower than at the lowest point.	

[Turn over

10	Two points P and Q are located a fixed distance apart on a straight line joining them to an object